Game Theory
Correlated equilibrium[1]

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Correlated equilibrium I

Example (correlated equilibrium 1)

	L	R
U	5,1	0,0
D	4,4	1,5

Table: correlated equilibrium

Find the three Nash equilibria!

Correlated equilibrium II

Example (correlated equilibrium 1)

	L	R
U	5,1	0,0
D	4,4	1,5

Table: correlated equilibrium

- pure strategy NE give high aggregate but very unequal payoff
- mixed strategy equilibrium gives equal but low payoff

Correlated equilibrium III

- Can players get equal and high payoffs?
- flip a coin: if tails (U, L), if head (D, R)
- with "unfair coins" any payoff in the convex hull of the NE payoffs is attainable
- can players do even better?
- Yes:
  - randomization device with three equally likely states A,B and C
  - P1 gets a message iff state is A
  - P2 gets a message iff state is C

  Then the following is an equilibrium
  - P1 plays U when he gets a message and D otherwise
  - P2 plays R when he gets a message and L otherwise

  Check that no deviation is profitable!

Correlated equilibrium IV

- expected payoff
  $1/3(5,1) + 1/3(4,4) + 1/3(1,5) = (3.33, 3.33)$ is outside of the convex hull of the Nash payoffs
- correlated equilibrium can lead to higher payoffs than NE

- interpretation correlated equilibrium:
  - both players first communicate and *construct a correlation machine together*
    each player sees output of the machine before taking action
  - *impartial mediator* gives (privately!) recommendations a_i to each player according to some probability distribution
    Recommendations are self-enforcing

Correlated equilibrium V

We now use the second interpretation:

- take a strategic form game $G = \langle N, (A_i), (u_i) \rangle$
- a probability distribution p over A leads to the game $G^*(p)$:
  1. mediator draws an action profile $a = (a_1, \ldots, a_n)$ from A according to probability distribution p
  2. mediator reveals a_i to each player i (but does not reveal a_{-i})
  3. each player chooses an action $a'_i \in A_i$
  4. payoff for each player i is $u_i(a'_1, \ldots, a'_n)$
- pure strategy for player i in $G^*(p)$ is function $s_i : A_i \to A_i$ (action as function of recommendation)
- belief of player i when getting recommendation a_i:

$$p(a_{-i}|a_i) = \frac{p(a_i, a_{-i})}{\sum_{b_{-i} \in A_{-i}} p(a_i, b_{-i})}$$

Correlated equilibrium VI

Lemma (MSZ Thm 8.5)

All players following the recommendation, i.e. $s_i(a_i) = a_i$ *for all players* i*, is an equilibrium of* $G^*(p)$ *if and only if*

$$\sum_{a_{-i} \in A_{-i}} p(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} p(a_i, a_{-i}) u_i(a_i', a_{-i}) \quad (1)$$

for all players i *and all actions* $a_i, a_i' \in A_i$.

Correlated equilibrium VII

Proof. Expected utility of player i when a_i after receiving recommendation a_i is

$$EU_i(a_i) = \sum_{a_{-i} \in A_{-i}} \frac{p(a_i, a_{-i})}{\sum_{b_{-i} \in A_{-i}} p(a_i, b_{-i})} u_i(a_i, a_{-i}).$$

Expected utility of playing a_i' after receiving recommendation a_i' is

$$EU_i(a_i') = \sum_{a_{-i} \in A_{-i}} \frac{p(a_i, a_{-i})}{\sum_{b_{-i} \in A_{-i}} p(a_i, b_{-i})} u_i(a_i', a_{-i}).$$

$EU_i(a_i) \geq EU_i(a_i')$ if and only if (1) holds. $\square$

For this proof, we use the convention $\frac{p(a_i, a_{-i})}{\sum_{b_{-i} \in A_{-i}} p(a_i, b_{-i})} = 0$ if $p(a_i, b_{-i}) = 0$ for all $b_{-i} \in A_{-i}$.

Correlated equilibrium VIII

Definition (correlated equilibrium)

A probability distribution p over A is a correlated equilibrium in the strategic form game $G = \langle N, (A_i), (u_i) \rangle$ if $s_i(a_i) = a_i$ for all players i is an equilibrium of $G^*(p)$.

Correlated equilibrium IX

Proposition

Let α^* be a mixed strategy equilibrium. Then the distribution p_{α^*} defined by

$$p_{\alpha^*}(a_1, \ldots, a_n) = \Pi_{i=1}^{n} \alpha_i^*(a_i)$$

is a correlated equilibrium.

Proof.

Correlated equilibrium X

Corollary

A correlated equilibrium exists in all finite games.

Correlated equilibrium XI

Example (correlated equilibrium 2)

	L	R
U	0,1,3	0,0,0
D	1,1,1	1,0,0

A

	L	R
U	2,2,2	0,0,0
D	2,2,0	2,2,2

B

	L	R
U	0,1,0	0,0,0
D	1,1,0	1,0,3

C

Table: correlated equilibrium 2

Correlated equilibrium XII

Example (correlated equilibrium 2 continued)

- unique NE is (D,L,A) (check!)
- getting the $(2, 2, 2)$ payoff from (U, L, B) or from (D, R, B) would be nice for all players but P3 has an incentive to deviate (either to A or to C)
- this example: limiting your own information can be beneficial
- what could a mediator do to solve this problem?

Correlated equilibrium XIII

Proposition (MSZ Thm 8.9)

Let $G = \langle N, (A_i), (u_i) \rangle$ be a strategic form game. The set of correlated equilibria of G is convex.
That is, if $p' \in \Delta A$ and $p'' \in \Delta A$ are correlated equilibria, then $p = \alpha p' + (1 - \alpha) p''$ is a correlated equilibrium for any $\alpha \in [0, 1]$.

Proof.

Review Questions

- What is a correlated equilibrium and how can it be interpreted?
- Why can correlating recommendations (sometimes) help players to achieve a higher payoff than in any Nash equilibrium?
- Explain why every mixed strategy Nash equilibrium can be interpreted as a correlated equilibrium.

reading: MSZ ch. 8
*reading: Jann and Schottmüller "Correlated equilibria in homogenous good Bertrand competition", *Journal of Mathematical Economics*, Vol. 57, March 2015, pp. 31-37

Exercises I

1. Determine all pure and mixed Nash equilibria of the following game. Find a correlated equilibrium in which the sum of the players' payoff is higher than in any Nash equilibrium.

	L	R
U	0,0	6,2
D	2,6	5,5

2. repeat the previous exercise with the following game

	L	C	R
U	1,1	2,4	4,2
M	4,2	1,1	2,4
D	2,4	4,2	1,1

3. Show that all the actions that are played with positive probability in a correlated equilibrium are rationalizable.

Exercises II

4. *Think of a Bertrand game (2 firms with zero costs set price; one consumer buys from the firm with lowest price if this price is less than his valuation 1\$). Assume that prices have to be in whole cents. Show that there is no correlated equilibrium that leads to higher total payoffs than the pure strategy Nash equilibrium $(0.02, 0.02)$.
   Hint: Think of the highest recommendation given with positive probability in a correlated equilibrium.